Table 3: Downside Risk Factor Screening

		Horizon			
Model	Class	1	3	6	12
MUNC	Macro	80.8	82.5	76.7	70.0
PCF1	Stat	70.0	66.7	57.5	47.5
FUNC	Fin	62.5	65.0	60.8	50.0
QF1	Stat	70.0	52.5	45.0	25.0
NFCI	Fin	43.3	49.2	52.5	45.8
PC1	Stat	50.8	58.3	48.3	32.5
VIX	Fin	58.3	53.3	45.0	33.3
CISS	Fin	50.0	48.3	43.3	19.2
EBP	Fin	35.8	43.3	38.3	35.8
EPU	Text	35.0	32.5	23.3	14.2
CTG	Macro	25.8	19.2	22.5	28.3
RISK	Text	20.8	29.2	10.0	6.7
HPI	Macro	10.8	23.3	19.2	12.5
PRISK	Text	16.7	18.3	8.3	18.3
NPRISK	Text	16.7	14.2	10.0	11.7
WUI	Text	18.3	18.3	8.3	6.7
GPR	Text	4.2	0.8	6.7	10.0

This table reports the results of the downside risk factor screening exercise. For each predictor, the table lists the percentage of macroeconomic variables for which the coefficient on the predictor is statistically significant at the 5% level. The percentages are obtained from quantile regression, with $p = 0.05$, at horizons of 1, 3, 6, and 12 months. Significance is assessed on the basis of bootstrap standard errors. Predictors are sorted in descending order according to the average percentage of significant coefficients across horizons.

Table 4: Downside Risk Model Screening

	FRED-MD		$\overline{TLG}$			
h	Class	Model	QR	LR	SR	LSR
1	Bench	HIST	0.0	0.0	0.0	0.0
		AR	14.6	22.6	22.6	22.6
	Fin	NFCI	16.9	22.6	23.0	23.1
		EBP	16.3	22.4	22.6	22.7
		FUNC	18.2	23.3	24.2	24.2
		VIX	17.2	22.9	23.2	23.2
		CISS	19.1	23.7	24.2	24.2
	Macro	MUNC	20.9	23.1	25.2	**25.4**
		HPI	15.8	23.1	22.9	23.0
		CTG	15.6	22.6	22.7	22.7
		PC1	17.3	23.5	23.1	23.9
		QF1	19.1	23.1	23.7	23.9
		PCF1	20.6	23.7	25.0	25.0
		WUI	15.5	22.6	22.7	22.7
		GPR	14.8	22.6	22.6	22.7
		EPU	16.7	22.7	23.1	23.1
3	Bench	HIST	0.0	0.0	0.0	0.0
		AR	16.2	22.1	22.1	22.1
	Fin	NFCI	20.7	22.7	23.1	24.0
		EBP	18.9	21.7	21.5	22.2
		FUNC	23.2	24.2	26.3	27.2
		VIX	20.7	23.2	23.7	24.0
		CISS	21.6	24.0	23.9	24.6
	Macro	MUNC	26.1	23.5	26.8	**27.6**
		HPI	18.2	22.9	22.7	23.1
		CTG	17.6	22.2	22.3	22.4
		PC1	20.5	24.1	23.4	25.0
		QF1	22.3	23.1	24.5	24.8
		PCF1	24.5	24.3	25.3	25.9
		WUI	17.9	22.3	22.9	23.0
		GPR	16.6	22.2	22.1	22.3
		EPU	19.0	22.3	22.7	22.8
6	Bench	HIST	0.0	0.0	0.0	0.0
		AR	15.3	19.2	19.2	19.2
	Fin	NFCI	21.8	21.3	22.0	23.7
		EBP	19.5	20.0	20.2	21.7
		FUNC	24.0	22.4	24.9	25.9
		VIX	20.2	20.7	21.4	21.9
		CISS	22.0	22.2	22.0	23.0
	Macro	MUNC	**26.4**	21.3	24.2	25.1
		HPI	19.0	21.4	21.0	21.6
		CTG	17.6	19.5	20.0	20.2
		PC1	19.8	21.6	21.5	23.3
		QF1	20.6	20.3	21.5	21.9
		PCF1	25.6	23.1	24.6	25.3
		WUI	17.2	19.4	19.7	20.2
		GPR	15.7	19.2	19.2	19.4
		EPU	18.0	19.5	19.2	19.4
12	Bench	HIST	0.0	0.0	0.0	0.0
		AR	11.1	14.7	14.7	14.7
	Fin	NFCI	16.0	15.8	15.5	17.2
		EBP	14.7	16.4	15.7	17.6
		FUNC	18.3	17.7	17.9	20.4
		VIX	15.2	16.0	16.4	16.8
		CISS	17.4	18.0	18.6	19.1
	Macro	MUNC	**21.6**	16.6	17.1	19.2
		HPI	14.9	17.7	17.1	18.4
		CTG	15.1	15.9	16.8	17.6
		PC1	14.3	16.3	16.0	17.4
		QF1	14.5	15.3	15.5	15.8
		PCF1	21.2	19.1	19.5	21.0
		WUI	13.4	15.3	15.3	16.1
		GPR	11.9	14.8	15.2	15.4
		EPU	14.1	15.2	14.9	14.9

This table reports the results of the downside risk model screening exercise. For each downside risk factor and forecast horizon, the table lists the average tick loss gain associated with quantile regression, with $p = 0.05$, and the three variants of the location-scale regression at horizons of 1, 3, 6, and 12 months. The gains are computed relative to the historical quantile benchmark. Bold values indicate the best-performing specification at each horizon in terms of the average tick loss gain.

Table 5: Model Confidence Set

	Horizon							
	1		3		6		12	
	%	TLG	%	TLG	%	TLG	%	TLG
All	38.5	26.9	49.1	29.0	70.0	27.9	59.7	19.4
QR	0.0	22.1	0.7	24.5	46.1	24.9	42.6	17.6
LSR	77.3	26.9	97.9	29.0	94.3	27.9	77.3	19.4
1 Factor	42.9	26.6	50.0	28.7	60.7	27.9	71.4	19.4
2 Factors	43.6	26.9	51.3	29.0	79.5	27.1	70.5	18.3
3 Factors	35.6	26.9	48.3	29.0	67.2	27.0	52.9	17.2
QR NFCI	×	16.8	×	19.5	✓	18.4	✓	9.1
LSR NFCI	✓	25	✓	25.4	✓	23.5	✓	14.3
QAR(1)	×	14	×	15.9	✓	15.6	✓	11.1
AR(1)-GARCH(1,1)	✓	24.1	✓	24	✓	20.9	✓	16.2

This table reports the results of the Model Confidence Set procedure for the downside risk forecasts at horizons 1, 3, 6, and 12. For each forecast horizon, the table lists the share of models included in the MCS and the highest average tick loss gain within each specification subgroup. The first row refers to the full set of models, while the following rows decompose the results by model type and by number of factors. The bottom rows report, for selected individual models, whether the specification is included in the MCS together with the corresponding average tick loss gain.

Table 6: Downside Risk Prediction Using One Factor

	FRED-MD		QR						LSR					
h	Class	Model	$\overline{TLG}$	W	$\overline{H}$	DQU	DQC	MCS	$\overline{TLG}$	W	$\overline{H}$	DQU	DQC	MCS
1	Bench	HIST	0.0	1	6.4	47.5	26.7	×	0.0	1	6.4	47.5	26.7	×
		AR	14.0	0	6.2	47.5	36.7	×	24.1	0	5.3	76.7	80.8	✓
	Fin	NFCI	16.8	1	7.3	35.0	40.8	×	25.0	5	6.3	57.5	65.0	✓
		EBP	16.4	0	6.3	47.5	47.5	×	24.9	4	5.6	70.0	83.3	✓
		FUNC	17.4	2	6.1	49.2	48.3	×	25.5	7	5.3	71.7	81.7	✓
		VIX	16.2	0	6.1	47.5	50.8	×	24.1	1	5.4	74.2	79.2	✓
		CISS	18.4	2	6.9	44.2	49.2	×	25.1	9	6.1	68.3	74.2	✓
	Macro	MUNC	21.0	6	6.2	45.8	53.3	×	**26.6**	**41**	5.6	73.3	74.2	✓
		HPI	14.1	0	6.3	54.2	43.3	×	23.4	7	5.5	78.3	80.8	×
		CTG	13.5	0	7.0	50.8	39.2	×	23.7	2	5.8	72.5	75.0	✓
		PC1	15.1	0	5.8	47.5	50.0	×	23.9	3	5.3	70.0	81.7	✓
		QF1	18.5	4	5.8	50.8	54.2	×	24.8	6	5.3	75.8	80.8	✓
		PCF1	19.6	1	6.6	51.7	46.7	×	25.7	10	5.9	70.8	72.5	✓
		WUI	14.1	0	5.8	55.0	38.3	×	23.7	1	5.2	78.3	82.5	✓
		GPR	13.1	0	6.4	48.3	35.8	×	23.8	2	5.5	74.2	80.0	✓
		EPU	14.3	0	5.4	55.8	47.5	×	23.0	5	4.9	71.7	84.2	×
3	Bench	HIST	0.0	0	6.7	54.2	30.0	×	0.0	0	6.7	54.2	30.0	×
		AR	15.9	0	6.1	58.3	53.3	×	24.0	0	5.6	83.3	80.0	✓
	Fin	NFCI	19.5	3	8.5	25.8	37.5	×	25.4	3	7.4	45.8	55.0	✓
		EBP	19.4	1	6.1	56.7	52.5	×	25.6	2	5.7	78.3	75.0	✓
		FUNC	22.1	2	6.5	54.2	55.0	×	28.4	21	5.8	72.5	71.7	✓
		VIX	18.7	1	6.2	50.0	55.8	×	24.8	3	5.6	79.2	77.5	✓
		CISS	21.1	1	6.9	60.8	55.0	×	25.9	3	6.3	72.5	77.5	✓
	Macro	MUNC	25.5	15	6.6	47.5	51.7	×	**28.7**	**25**	6.1	67.5	65.0	✓
		HPI	15.9	0	6.5	63.3	47.5	×	23.6	3	6.1	79.2	75.0	✓
		CTG	14.7	0	7.5	55.8	47.5	×	23.4	11	6.3	70.8	74.2	✓
		PC1	17.6	0	5.9	53.3	54.2	×	25.0	4	5.4	74.2	80.0	✓
		QF1	21.1	0	5.7	50.8	63.3	×	26.0	3	5.5	74.2	75.8	✓
		PCF1	23.4	3	7.0	59.2	56.7	×	26.6	8	6.4	67.5	70.0	✓
		WUI	15.5	1	5.9	64.2	56.7	×	23.5	2	5.5	80.8	73.3	✓
		GPR	14.9	0	6.4	58.3	50.0	×	23.6	2	5.8	84.2	77.5	✓
		EPU	16.4	1	5.0	72.5	66.7	×	22.5	2	5.0	84.2	75.8	✓
6	Bench	HIST	0.0	1	6.9	63.3	50.8	×	0.0	1	6.9	63.3	50.8	×
		AR	15.6	0	6.3	64.2	63.3	✓	20.9	2	5.8	78.3	87.5	✓
	Fin	NFCI	18.4	0	9.4	35.8	55.8	✓	23.5	11	8.2	51.7	70.8	✓
		EBP	19.6	1	6.5	61.7	74.2	×	24.7	4	5.7	77.5	87.5	✓
		FUNC	22.4	4	7.0	57.5	58.3	✓	**27.9**	16	6.1	73.3	85.0	✓
		VIX	18.5	0	6.5	54.2	64.2	×	23.2	1	5.7	78.3	87.5	✓
		CISS	18.8	5	7.5	64.2	68.3	✓	23.4	6	6.9	76.7	84.2	✓
	Macro	MUNC	24.7	**24**	7.4	42.5	60.0	✓	25.9	8	6.5	59.2	71.7	✓
		HPI	14.4	0	7.2	66.7	67.5	×	21.4	3	6.5	85.0	85.0	✓
		CTG	13.8	1	8.3	57.5	59.2	×	20.2	5	6.9	71.7	80.0	✓
		PC1	17.4	3	6.2	63.3	56.7	×	23.6	6	5.8	79.2	84.2	✓
		QF1	20.8	2	5.9	58.3	57.5	×	24.0	1	5.5	76.7	83.3	✓
		PCF1	21.2	5	7.8	60.0	63.3	✓	24.6	2	6.8	74.2	85.8	✓
		WUI	13.9	2	6.4	67.5	65.8	×	18.9	0	6.0	85.0	83.3	×
		GPR	13.9	0	6.7	64.2	56.7	×	19.8	2	6.0	78.3	81.7	✓
		EPU	15.2	1	5.3	71.7	67.5	×	18.6	4	5.1	81.7	77.5	×
12	Bench	HIST	0.0	1	7.5	68.3	64.2	×	0.0	1	7.5	68.3	64.2	×
		AR	11.1	0	6.8	72.5	78.3	✓	16.2	3	6.3	81.7	78.3	✓
	Fin	NFCI	9.1	4	10.0	54.2	75.0	✓	14.3	8	9.3	62.5	83.3	✓
		EBP	13.1	10	6.8	72.5	80.8	✓	18.8	13	6.8	80.8	90.0	✓
		FUNC	13.9	1	8.0	69.2	80.0	✓	**19.4**	5	7.1	79.2	75.8	✓
		VIX	13.2	1	7.0	68.3	77.5	✓	17.5	1	6.4	75.8	87.5	✓
		CISS	9.4	1	8.4	77.5	80.8	✓	15.7	3	7.9	84.2	86.7	✓
	Macro	MUNC	17.6	**21**	8.4	56.7	74.2	✓	16.7	11	7.7	69.2	77.5	✓
		HPI	5.9	1	7.8	72.5	82.5	×	12.7	7	7.4	75.8	85.8	✓
		CTG	3.5	3	10.0	56.7	76.7	×	12.4	6	8.2	72.5	76.7	✓
		PC1	9.7	2	7.0	70.0	76.7	✓	16.6	5	6.6	76.7	82.5	✓
		QF1	12.9	1	6.8	71.7	79.2	✓	16.4	1	6.5	80.8	80.8	✓
		PCF1	11.5	0	9.3	63.3	80.8	✓	16.7	7	8.3	80.0	85.0	✓
		WUI	7.6	0	7.3	74.2	78.3	×	12.8	2	6.9	80.8	84.2	×
		GPR	9.4	0	7.1	70.8	84.2	×	14.7	0	6.6	80.8	80.8	×
		EPU	8.4	1	6.0	81.7	85.0	×	12.2	1	5.8	83.3	83.3	×

This table reports backtesting statistics for the downside risk forecasts of one-factor specifications at horizons 1, 3, 6, and 12. For each forecast horizon and specification, the table lists the average tick loss gain, the number of pairwise wins against other one-factor models, the grand average of the hit variable, the fraction of series for which the null of correct coverage in the DQU and DQC tests cannot be rejected at the 5% level, and whether the specification is included in the model confidence set. Bold values indicate the best-performing specifications at each horizon in terms of the average tick loss gain or the number of pairwise wins.

Table 7: Downside Risk Prediction Using Multiple Factors

FRED-MD			QR					LSR				
h	Class	Model	$\overline{TLG}$	$\overline{H}$	DQU	DQC	MCS	$\overline{TLG}$	$\overline{H}$	DQU	DQC	MCS
1	Bench	HIST	0.0	6.4	47.5	26.7	×	0.0	6.4	47.5	26.7	×
	Macro	MUNC	21.0	6.2	45.8	53.3	×	26.6	5.6	73.3	74.2	✓
	Macro + Fin	MUNC + FUNC	20.7	6.3	51.7	54.2	×	26.8	5.6	70.0	71.7	✓
		MUNC + NFCI	22.1	6.6	45.0	49.2	×	**26.9**	6.0	61.7	70.8	✓
		MUNC + EBP	20.9	6.8	44.2	43.3	×	26.8	6.0	68.3	69.2	✓
		MUNC + FUNC + EBP	20.6	6.8	44.2	42.5	×	26.4	6.0	65.0	64.2	✓
	Macro + Text	MUNC + EPU	19.6	6.4	50.0	50.0	×	25.6	5.6	73.3	80.0	✓
	Macro + Stat	MUNC + PC1	20.6	6.1	52.5	50.8	×	26.1	5.4	67.5	72.5	✓
		MUNC + PC1 + QF1	20.4	6.2	55.8	52.5	×	25.2	5.5	70.8	75.8	✓
	Fin	FUNC + CISS	18.8	6.7	50.8	47.5	×	25.5	5.9	71.7	74.2	✓
		FUNC + EBP	17.4	6.4	50.8	48.3	×	25.4	5.7	72.5	79.2	✓
		CISS + EBP	17.3	7.5	38.3	41.7	×	24.8	6.3	63.3	73.3	✓
	Fin + Text	FUNC + EPU	16.6	5.9	58.3	55.0	×	24.3	5.2	73.3	80.8	×
	Fin + Stat	NFCI + PC1	16.8	6.9	41.7	40.8	×	24.6	5.9	64.2	74.2	✓
	Fin + Stat + Text	FUNC + PC1 + EPU	16.5	5.9	56.7	55.8	×	23.5	5.3	68.3	77.5	×
		FUNC + QF1 + EPU	18.2	6.0	52.5	58.3	×	24.1	5.3	72.5	80.8	×
		NFCI + CISS + QF1	19.1	6.7	42.5	48.3	×	24.3	6.2	60.0	65.8	✓
		NFCI + EPU	17.6	6.8	45.0	40.0	×	24.6	6.1	62.5	65.8	✓
	Stat	QF1 + QF2 + QF3	18.5	6.1	53.3	56.7	×	24.6	5.3	75.8	80.8	✓
		PC1 + QF1	18.2	5.8	53.3	59.2	×	24.1	5.2	71.7	75.0	✓
	Stat + Text	PC1 + QF1 + EPU	17.5	5.6	58.3	65.8	×	22.4	5.2	78.3	83.3	×
3	Bench	HIST	0.0	6.7	54.2	30.0	×	0.0	6.7	54.2	30.0	×
	Macro	MUNC	25.5	6.6	47.5	51.7	×	28.7	6.1	67.5	65.0	✓
	Macro + Fin	MUNC + FUNC	25.0	7.0	51.7	54.2	×	**29.0**	6.3	65.0	66.7	✓
		MUNC + NFCI	24.5	7.6	42.5	40.8	✓	27.2	7.1	48.3	59.2	✓
		MUNC + EBP	24.3	7.6	46.7	45.8	×	28.2	6.7	62.5	60.8	✓
		MUNC + FUNC + EBP	23.5	7.8	42.5	40.0	×	28.0	6.8	60.0	64.2	✓
	Macro + Text	MUNC + EPU	23.5	6.8	59.2	53.3	×	26.9	6.2	69.2	67.5	✓
	Macro + Stat	MUNC + PC1	25.3	6.5	50.0	56.7	×	28.2	5.9	63.3	67.5	✓
		MUNC + PC1 + QF1	25.2	6.6	50.8	54.2	×	28.3	6.0	65.8	68.3	✓
	Fin	FUNC + CISS	22.3	6.8	58.3	58.3	×	27.5	6.3	77.5	74.2	✓
		FUNC + EBP	20.6	7.1	50.8	46.7	×	27.4	6.0	70.8	70.8	✓
		CISS + EBP	20.0	7.5	48.3	43.3	×	25.5	6.7	63.3	68.3	✓
	Fin + Text	FUNC + EPU	21.8	6.1	64.2	60.0	×	26.6	5.7	75.8	74.2	✓
	Fin + Stat	NFCI + PC1	19.7	7.9	38.3	45.0	×	25.4	6.9	50.8	64.2	✓
	Fin + Stat + Text	FUNC + PC1 + EPU	21.1	6.2	60.8	60.0	×	26.2	5.7	76.7	76.7	✓
		FUNC + QF1 + EPU	21.9	6.3	62.5	64.2	×	26.3	5.8	74.2	77.5	✓
		NFCI + CISS + QF1	20.8	7.7	43.3	45.0	×	24.4	7.0	56.7	63.3	✓
		NFCI + EPU	20.0	7.9	42.5	46.7	×	24.5	7.1	53.3	57.5	✓
	Stat	QF1 + QF2 + QF3	20.1	6.1	52.5	58.3	×	25.4	5.5	70.8	80.0	✓
		PC1 + QF1	20.2	5.7	49.2	66.7	×	24.7	5.5	74.2	79.2	✓
	Stat + Text	PC1 + QF1 + EPU	20.1	5.5	63.3	67.5	×	23.0	5.2	78.3	79.2	✓
6	Bench	HIST	0.0	6.9	63.3	50.8	×	0.0	6.9	63.3	50.8	×
	Macro	MUNC	24.7	7.4	42.5	60.0	✓	25.9	6.5	59.2	71.7	✓
	Macro + Fin	MUNC + FUNC	23.8	8.0	45.0	48.3	✓	**27.0**	6.8	61.7	70.0	✓
		MUNC + NFCI	22.0	9.0	38.3	58.3	✓	22.7	8.2	45.8	65.8	✓
		MUNC + EBP	22.9	8.1	52.5	60.0	✓	23.8	7.0	63.3	78.3	✓
		MUNC + FUNC + EBP	21.4	8.8	45.8	55.8	✓	23.4	7.5	55.8	70.8	✓
	Macro + Text	MUNC + EPU	22.2	8.0	50.0	60.0	✓	23.7	6.7	70.0	73.3	✓
	Macro + Stat	MUNC + PC1	24.9	7.3	50.8	64.2	✓	24.9	6.4	65.8	78.3	✓
		MUNC + PC1 + QF1	24.6	7.4	54.2	60.0	✓	26.3	6.6	65.0	73.3	✓
	Fin	FUNC + CISS	21.4	7.6	61.7	66.7	✓	25.1	6.8	79.2	86.7	✓
		FUNC + EBP	20.6	8.0	51.7	63.3	✓	25.2	6.5	70.8	82.5	✓
		CISS + EBP	18.2	8.3	49.2	69.2	×	22.8	7.3	71.7	78.3	✓
	Fin + Text	FUNC + EPU	21.3	6.8	68.3	68.3	✓	25.7	6.0	76.7	81.7	✓
	Fin + Stat	NFCI + PC1	18.1	9.1	39.2	58.3	✓	23.6	7.9	55.8	72.5	✓
	Fin + Stat + Text	FUNC + PC1 + EPU	20.6	7.0	63.3	65.0	✓	25.3	6.1	76.7	77.5	✓
		FUNC + QF1 + EPU	21.4	7.0	64.2	65.8	✓	25.2	6.1	77.5	75.8	✓
		NFCI + CISS + QF1	17.7	8.8	44.2	62.5	×	21.2	7.9	60.8	71.7	✓
		NFCI + EPU	18.0	9.3	36.7	53.3	✓	22.3	8.3	42.5	75.0	✓
	Stat	QF1 + QF2 + QF3	19.7	6.3	64.2	65.0	×	23.2	5.4	75.8	80.0	✓
		PC1 + QF1	19.2	6.0	60.0	59.2	×	23.7	5.8	79.2	81.7	✓
	Stat + Text	PC1 + QF1 + EPU	18.1	5.8	64.2	68.3	×	21.8	5.4	76.7	79.2	✓
12	Bench	HIST	0.0	7.5	68.3	64.2	×	0.0	7.5	68.3	64.2	×
	Macro	MUNC	17.6	8.4	56.7	74.2	✓	16.7	7.7	69.2	77.5	✓
	Macro + Fin	MUNC + FUNC	14.1	9.2	59.2	70.0	✓	16.4	8.5	66.7	74.2	✓
		MUNC + NFCI	13.9	9.5	56.7	79.2	✓	12.4	9.3	55.0	75.8	✓
		MUNC + EBP	14.0	9.3	64.2	79.2	✓	14.2	8.5	68.3	74.2	✓
		MUNC + FUNC + EBP	10.6	10.2	61.7	79.2	×	12.6	9.0	67.5	78.3	✓
	Macro + Text	MUNC + EPU	14.1	9.4	55.0	77.5	×	14.1	8.3	73.3	80.0	✓
	Macro + Stat	MUNC + PC1	16.7	8.7	58.3	68.3	✓	15.9	8.0	66.7	78.3	✓
		MUNC + PC1 + QF1	16.5	8.9	52.5	70.0	✓	17.2	8.4	64.2	75.8	✓
	Fin	FUNC + CISS	9.1	9.0	72.5	88.3	×	13.7	8.7	81.7	86.7	✓
		FUNC + EBP	9.7	8.9	68.3	77.5	×	**17.6**	7.4	75.8	85.8	✓
		CISS + EBP	10.5	9.5	60.8	73.3	×	15.5	8.9	74.2	79.2	✓
	Fin + Text	FUNC + EPU	10.8	7.9	75.0	78.3	×	15.8	7.0	80.8	74.2	×
	Fin + Stat	NFCI + PC1	7.7	10.3	50.8	83.3	✓	14.2	9.3	60.0	83.3	✓
	Fin + Stat + Text	FUNC + PC1 + EPU	8.4	8.5	65.8	78.3	×	15.0	7.4	75.8	77.5	×
		FUNC + QF1 + EPU	10.3	8.2	70.0	74.2	×	14.5	7.2	80.8	78.3	×
		NFCI + CISS + QF1	4.2	10.3	59.2	76.7	✓	10.1	9.7	70.0	80.0	✓
		NFCI + EPU	7.0	10.5	44.2	72.5	×	12.4	9.6	58.3	73.3	✓
	Stat	QF1 + QF2 + QF3	11.7	7.2	70.0	75.8	✓	15.3	6.5	75.0	81.7	✓
		PC1 + QF1	10.6	7.0	70.0	76.7	✓	16.2	6.7	76.7	84.2	✓
	Stat + Text	PC1 + QF1 + EPU	8.4	6.9	75.8	79.2	×	13.3	6.3	80.0	80.8	×

This table reports backtesting statistics for the downside risk forecasts of selected one-, two-, and three-factor specifications at horizons 1, 3, 6, and 12. For each forecast horizon and specification, the table lists the average tick loss gain, the grand average of the hit variable, the fraction of series for which the null of correct coverage in the DQU and DQC tests cannot be rejected at the 5% level, and whether the specification is

Table 8: Downside Risk Prediction for the FRED-MD Categories

h	Class	Model	Output	Labor	Hous	Ord Inv	Credit	Exch Rates	Prices Down	Prices Up	Stocks
1	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	5.1	17.7	72.2	8.7	4.4	7.9	2.9	3.3	19.3
		AR LS	14.5	26.5	74.4	21.2	15.6	12.6	15.0	18.2	28.1
	QR	NFCI	9.0	22.1	72.0	10.2	4.6	8.7	5.9	5.2	25.9
		EBP	6.7	21.1	72.0	13.3	4.1	6.0	6.3	6.5	21.7
		FUNC	10.7	22.1	72.5	15.6	4.6	7.3	6.9	4.3	28.2
		MUNC	16.0	25.0	72.4	20.1	5.3	7.2	11.5	10.8	25.6
		QF1	11.5	23.0	72.7	19.2	6.0	8.5	9.1	6.3	21.4
		PCF1	11.5	24.6	72.3	18.5	5.2	6.3	11.1	8.2	27.2
	LSR	NFCI	16.8	28.6	74.3	19.5	15.9	**12.8**	16.0	18.2	25.8
		EBP	14.4	28.6	74.4	22.2	15.4	12.4	16.2	18.1	29.0
		FUNC	17.3	28.2	74.6	23.2	15.0	11.2	17.2	18.0	**34.7**
		MUNC	**20.9**	29.6	74.4	**27.1**	14.6	8.5	17.8	**19.9**	27.9
		QF1	14.5	28.3	**74.8**	25.0	15.0	12.2	16.4	17.1	27.4
		PCF1	14.8	**30.0**	74.3	25.9	**15.9**	11.2	**17.8**	18.4	26.4
3	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	4.3	19.6	73.1	9.4	6.7	0.5	7.1	9.5	12.6
		AR LS	10.0	26.3	74.5	14.0	12.8	**7.6**	14.3	25.9	28.3
	QR	NFCI	12.3	24.7	73.3	15.0	6.7	0.1	8.8	12.2	18.1
		EBP	9.6	27.1	73.2	19.5	3.3	-5.2	7.7	12.2	17.4
		FUNC	18.4	29.6	74.4	23.7	7.3	-0.6	8.9	10.6	25.5
		MUNC	**24.0**	32.4	74.4	27.7	7.5	-0.2	13.4	16.6	20.0
		QF1	12.8	28.4	73.6	23.9	5.8	0.1	8.6	13.6	16.0
		PCF1	18.7	30.6	73.8	26.8	6.8	0.1	9.9	15.8	16.8
	LSR	NFCI	14.4	29.7	74.0	16.0	13.0	3.8	14.6	26.3	22.6
		EBP	13.9	31.6	75.1	19.7	11.3	2.9	14.5	24.6	24.1
		FUNC	20.1	32.6	**75.5**	23.5	12.9	7.2	17.3	**26.6**	**34.3**
		MUNC	22.1	**35.0**	75.0	**30.3**	12.9	6.1	**18.2**	23.0	22.9
		QF1	10.6	31.3	74.9	24.7	**13.0**	6.9	14.5	26.3	25.4
		PCF1	16.9	33.7	74.0	25.8	11.8	2.5	14.0	24.2	19.9
6	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	4.9	16.8	71.6	7.8	6.7	-1.0	9.1	11.7	8.3
		AR LS	4.1	21.3	73.0	10.4	10.4	**4.6**	13.0	24.3	**28.1**
	QR	NFCI	12.2	19.9	71.3	11.8	6.9	-4.9	11.1	13.2	16.4
		EBP	14.4	28.0	72.0	20.2	3.8	-8.7	7.6	10.0	13.8
		FUNC	17.8	29.9	73.3	24.4	8.6	-3.6	10.4	11.9	20.3
		MUNC	**21.9**	32.1	73.8	**27.4**	8.2	-5.4	12.5	16.6	17.6
		QF1	14.3	27.7	71.9	22.7	7.9	-1.8	9.4	12.5	11.2
		PCF1	15.2	29.2	73.1	20.8	6.3	-6.3	10.4	12.5	11.0
	LSR	NFCI	12.9	24.7	72.5	13.6	10.7	-1.4	14.7	27.2	22.2
		EBP	16.1	31.0	74.0	19.4	8.0	0.3	12.2	24.5	16.9
		FUNC	20.3	**33.5**	**75.5**	24.2	**11.0**	3.4	**14.9**	**27.3**	27.7
		MUNC	20.0	30.9	74.5	26.0	7.7	4.2	11.7	25.0	21.4
		QF1	12.6	28.6	73.3	19.7	9.1	2.9	12.0	25.2	21.1
		PCF1	13.7	31.0	73.1	20.6	10.2	1.2	12.8	25.5	9.5
12	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	1.5	11.4	66.6	2.2	7.5	0.2	3.5	6.1	6.0
		AR LS	1.0	12.8	68.2	-1.0	7.1	**3.4**	13.0	22.2	14.0
	QR	NFCI	1.5	5.9	63.1	0.7	3.2	-11.5	5.2	5.3	13.9
		EBP	6.8	22.4	66.9	13.7	2.0	-9.4	0.7	-1.2	11.4
		FUNC	4.3	18.6	67.4	**14.1**	7.7	-7.3	3.5	5.0	15.4
		MUNC	**10.0**	21.4	68.7	12.9	**8.4**	-5.9	6.8	13.4	16.0
		QF1	4.5	18.0	65.9	11.7	7.3	-3.9	3.1	3.0	7.9
		PCF1	-1.5	16.9	66.0	7.5	7.9	-15.0	2.3	6.1	2.5
	LSR	NFCI	1.9	8.3	65.0	3.2	1.5	-12.2	13.2	21.6	15.3
		EBP	4.9	**23.7**	67.7	10.2	-0.9	-9.4	**14.9**	18.5	11.5
		FUNC	5.8	22.6	**69.4**	12.2	1.8	1.5	11.2	21.6	**16.2**
		MUNC	9.2	17.7	68.5	12.9	-5.7	0.5	5.0	21.1	7.4
		QF1	4.9	17.6	67.9	7.3	0.2	-0.7	11.4	17.5	5.4
		PCF1	0.3	18.5	65.8	9.2	1.5	-6.5	11.5	**23.0**	-0.2

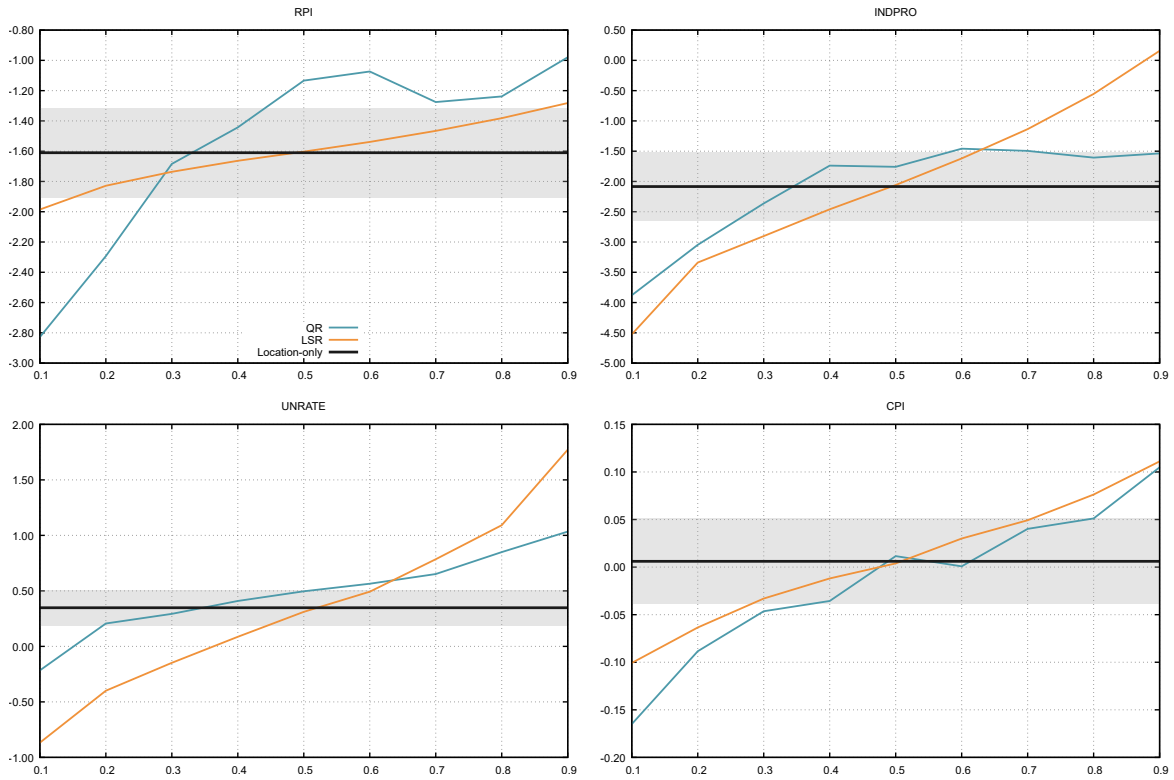
This table reports the out-of-sample predictive performance of selected one-factor downside risk models across the FRED–MD categories at horizons 1, 3, 6, and 12. For each forecast horizon, specification, and category, the table lists the average tick loss gain relative to the historical quantile benchmark. Results are reported separately for quantile regression and location-scale regression specifications. Bold values indicate the best-performing specification for each category at each horizon in terms of the average tick loss gain.

Table 9: Downside Risk Prediction for Policy-Relevant Variables

h	Class	Model	RPI	IP	UR	ALL EMP	HOU ST	PPI	PPI up	CPI	CPI up
1	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	-4.1	9.7	10.3	37.7	84.2	6.3	13.8	-0.0	-3.8
		AR LS	24.1	19.5	9.8	46.8	84.4	15.7	26.0	14.1	12.6
	QR	NFCI	-1.9	17.6	19.6	48.8	83.6	10.0	14.5	3.8	-2.9
		EBP	1.5	12.3	14.1	45.2	84.1	10.7	14.0	6.2	1.4
		FUNC	1.0	18.9	13.8	45.9	84.1	11.9	15.7	6.4	-2.0
		MUNC	5.6	**29.2**	20.4	50.5	84.5	18.7	22.6	14.3	10.8
		QF1	4.1	18.5	14.6	44.6	84.9	15.6	20.7	10.7	0.5
		PCF1	-5.0	21.9	16.6	**51.5**	83.4	15.2	18.3	13.1	2.1
	LSR	NFCI	27.1	22.0	17.8	50.5	84.1	16.2	26.3	14.1	11.5
		EBP	29.6	17.1	12.0	47.9	84.0	16.3	26.9	14.5	13.8
		FUNC	27.8	22.2	15.1	49.7	84.4	19.1	27.6	16.3	10.8
		MUNC	**30.8**	28.0	**22.1**	51.4	**85.3**	**19.4**	**31.9**	**18.6**	**15.6**
		QF1	26.2	15.8	15.1	48.0	85.2	16.2	31.1	16.7	8.2
		PCF1	15.5	19.9	17.9	49.5	84.4	19.2	28.3	15.6	12.6
3	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	-10.6	9.5	14.7	37.2	83.6	8.3	13.6	8.0	12.0
		AR LS	-3.1	20.5	19.9	39.3	83.7	9.3	33.8	12.2	31.3
	QR	NFCI	-1.0	22.7	32.0	44.1	83.7	8.8	14.6	8.1	13.4
		EBP	2.6	18.6	30.7	47.9	83.7	11.6	16.9	8.4	13.9
		FUNC	4.7	27.7	32.4	53.6	85.1	10.7	18.6	8.4	9.9
		MUNC	8.2	**39.6**	**35.0**	**57.5**	86.2	**15.9**	29.8	19.3	19.8
		QF1	9.7	22.0	25.4	48.0	84.3	12.0	20.4	8.0	18.0
		PCF1	4.6	30.6	32.8	51.6	84.5	9.3	19.6	9.4	18.9
	LSR	NFCI	-0.9	25.1	31.3	45.7	84.4	9.3	34.4	9.7	**31.7**
		EBP	11.1	21.1	26.1	50.9	85.2	10.5	32.5	12.9	28.3
		FUNC	4.5	29.3	30.1	53.8	85.9	12.6	**36.1**	13.0	31.2
		MUNC	**11.7**	34.7	34.1	55.1	**86.6**	12.9	34.0	**20.5**	26.6
		QF1	9.1	16.5	23.2	48.3	85.8	10.9	32.4	11.6	30.3
		PCF1	8.9	23.7	30.4	53.5	84.8	9.2	34.9	8.1	25.6
6	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	-6.9	8.7	15.9	25.7	81.0	11.9	28.8	9.4	13.2
		AR LS	-10.8	7.5	16.9	30.0	82.2	10.6	42.0	9.6	31.0
	QR	NFCI	-1.8	20.5	29.6	26.1	80.3	12.0	28.9	9.5	13.6
		EBP	12.4	21.0	36.2	47.0	82.2	11.0	29.4	4.2	5.8
		FUNC	16.6	27.4	34.2	49.4	83.1	11.7	29.9	9.4	11.5
		MUNC	16.1	**32.4**	**40.7**	**57.0**	86.2	12.0	38.1	**12.8**	19.2
		QF1	**18.0**	19.0	28.8	44.9	81.5	12.3	32.9	8.5	13.5
		PCF1	14.8	22.0	34.4	47.3	82.8	**12.5**	29.0	7.3	12.2
	LSR	NFCI	-1.8	20.1	27.2	36.8	80.2	11.8	39.3	9.6	**34.6**
		EBP	8.6	22.6	33.3	49.5	83.2	10.8	42.3	1.4	24.6
		FUNC	11.4	26.0	34.2	52.5	**86.4**	11.1	**45.3**	10.9	31.6
		MUNC	15.8	30.2	32.7	50.7	84.9	8.4	41.6	7.2	28.6
		QF1	13.7	18.1	26.3	43.8	83.2	10.6	43.1	4.9	27.3
		PCF1	11.9	16.4	31.9	48.3	83.0	11.3	43.7	4.4	22.4
12	Bench	HIST	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		AR QR	-1.9	4.2	12.7	12.1	73.9	-1.0	18.0	0.4	1.2
		AR LS	-4.9	4.5	9.6	19.2	74.4	4.2	41.0	8.5	20.8
	QR	NFCI	-1.8	6.2	19.7	-3.7	70.1	-0.5	14.2	-1.9	-2.2
		EBP	3.5	12.3	33.3	**41.0**	76.7	-0.9	16.5	-2.9	-7.9
		FUNC	6.2	11.0	22.2	25.0	74.9	-1.5	16.4	2.2	-2.9
		MUNC	17.3	**14.4**	**34.3**	35.3	**80.0**	1.6	32.6	5.2	12.1
		QF1	4.7	10.6	18.2	26.0	72.5	-0.1	17.5	2.1	0.1
		PCF1	2.3	-0.2	24.4	24.2	74.9	-0.2	16.2	1.4	-2.2
	LSR	NFCI	1.0	5.3	18.2	3.1	72.5	6.9	37.4	4.6	19.4
		EBP	10.2	7.6	30.8	29.8	74.8	**11.1**	37.4	11.7	17.9
		FUNC	3.8	3.3	15.8	36.7	79.9	3.1	**41.9**	8.7	**22.0**
		MUNC	**18.0**	12.9	17.4	28.3	77.2	-2.4	39.5	6.1	21.0
		QF1	10.0	9.5	16.5	28.0	75.9	3.1	38.0	10.1	8.9
		PCF1	2.5	3.2	28.0	24.9	75.4	5.9	39.7	**14.3**	17.3

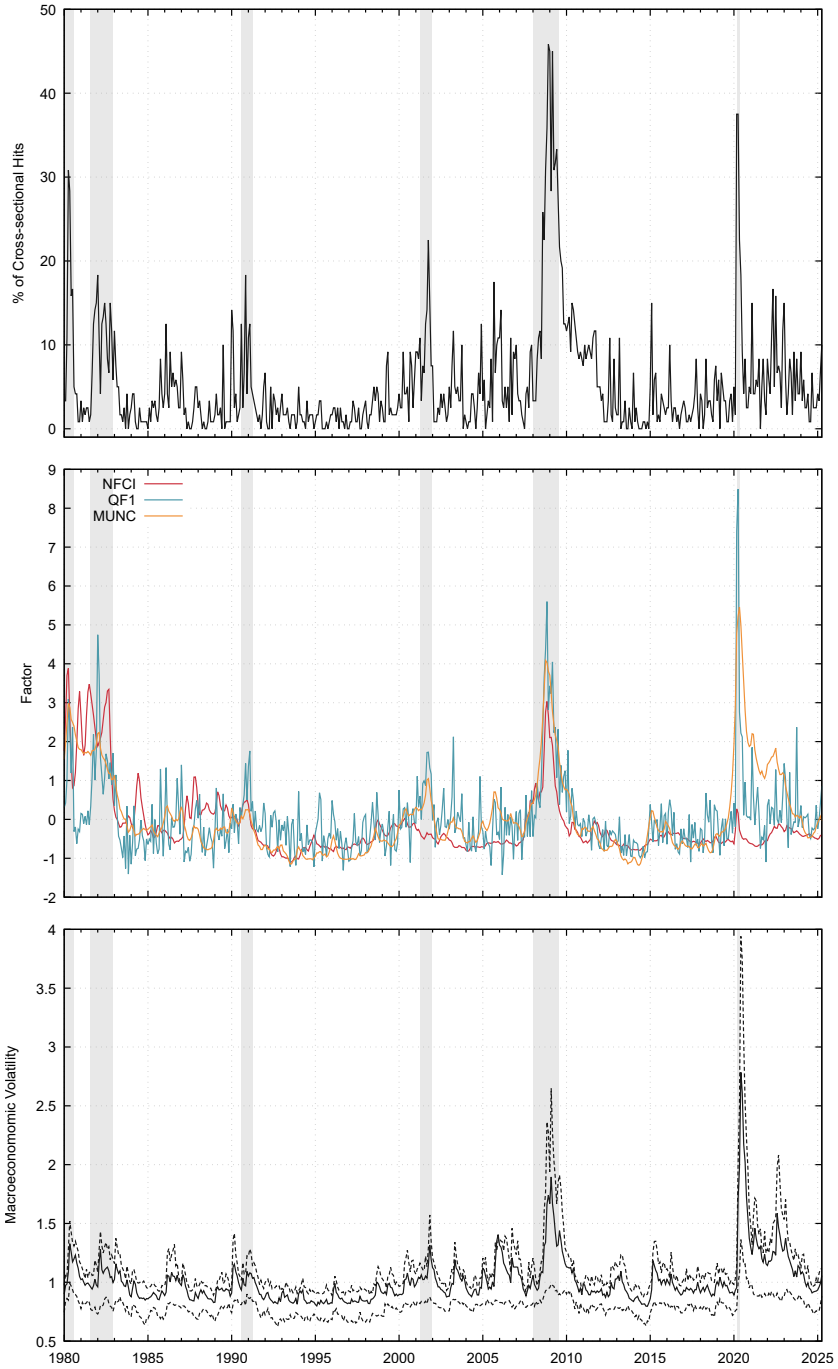
This table reports the out-of-sample predictive performance of one-factor downside risk models for selected policy-relevant variables at horizons 1, 3, 6, and 12. For each forecast horizon, specification, and variable, the table lists the average tick loss gain relative to the historical quantile benchmark. Results are reported separately for quantile regression and location-scale regression specifications. Bold values indicate the best-performing specification for each variable at each horizon in terms of the average tick loss gain.

Figure 1: Quantile Impulse Response Analysis



This figure displays the QIRFs of selected macroeconomic variables to an increase in MUNC from its median to its 95% quantile. The responses refer to the twelve-month forward average of Real Personal Income (RPI), Industrial Production (INDPRO), the Unemployment Rate (UNRATE), and the Consumer Price Index (CPI). The QIRFs are reported for nine equally-spaced quantiles from the 10% to the 90% of the distribution and are estimated using quantile regression and location–scale regression. The shaded area represents the 95% confidence band around the location-only QIRF.

Figure 2: Downside Risk Factors and Macroeconomic Volatility



This figure displays the relationship between selected downside risk factors and macroeconomic volatility. The top panel reports the share of macroeconomic variables with realizations below their 5% empirical quantile. The middle panel reports the time series of the three leading downside risk factors: NFCI, QF1, and MUNC. The bottom panel reports the cross-sectional average, along with the first and third quartiles, of the estimated macroeconomic volatility.